

CLINICAL STUDY PROTOCOL

Use of Acetylsalicylic Acid (ASA) for Enhanced Early Detection of Colorectal Neoplasms

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Participating sites are listed in an extra document.

TABLE OF CONTENTS

Abbreviations	8
1. SUMMARY	10
2. KEYWORDS	10
3. STUDY SYNOPSIS.....	11
3.1 Title	11
3.2 Objectives.....	11
3.3 Intervention	11
3.4 Endpoints.....	11
3.5 Study type.....	12
3.6 Study phase	12
3.7 Study population	12
3.8 Inclusion and exclusion criteria.....	12
3.9 Statistical analysis	13
3.10 Planned sample size	14
3.11 Study duration	14
4. BACKGROUND AND RATIONALE.....	15
4.1 Background	15
4.2 Benefit/ Risk assessment.....	16
5. STUDY DESIGN	18
6. DESIGN ASPECTS.....	22
6.1 Outcome measures	22
6.2 Inclusion / exclusion criteria	22
6.3 Controls / Comparators	24
6.4 Dose and route of administration	24
6.5 Randomization	24
6.6 Methods against bias	24
6.7 Analysis of stool samples.....	25
6.8 Analysis and storage of blood samples	25
6.9 Feasibility of recruitment	26
6.10 Stopping rules.....	26
6.11 Drop-outs.....	27
7. STATISTICAL APPROACH	28
7.1 Analysis	28
7.2 Proposed sample size / Power calculations	28
8. ETHICAL CONSIDERATIONS/ Risk-Benefit-Evaluation AND SAFETY.....	30
8.1 Potential risks of the study medication	30
8.2 Pharmacokinetics and pharmacodynamics of acetylsalicylic acid.....	30
8.3 Choice of acetylsalicylic acid dose	32
8.4 Potential benefits of the study medication	32
8.5 Safety.....	32
8.6 Emergency unblinding	33
8.7 Additional burden for the participants	33

8.8 Results of the stool tests	34
8.9 Results of optional biomarker studies	34
9. LEGAL CONSIDERATIONS	34
9.1 General legal information.....	34
9.2 Submission to the Ethics Committees and Competent Authority (BfArM).....	35
9.3 Notification of Authorities	35
9.4 Participant information and informed consent	35
9.5 Responsibilities	36
9.6 Insurance	36
9.7 Confidentiality.....	36
9.8 Archiving.....	36
9.9 Protocol modifications and amendments	37
9.10 Access to source data, quality control, data handling	37
9.11 Data collection.....	37
9.12 Patient medical records/source data	38
9.13 Drug accountability sheet.....	38
9.14 Funding.....	39
10. QUALITY ASSURANCE AND MONITORING.....	40
10.1 Documentation and data handling.....	40
10.2 Monitoring.....	40
10.3 Skills of the Research Physicians	41
10.4 Steering Committee.....	41
10.5 Publication policy	41
11. STUDY MEDICATION	42
11.1 General Information about Investigational Medicinal Product.....	42
11.2 Pharmacological properties	42
11.3 Pharmacokinetic properties	42
11.4 Information about potentially relevant adverse reactions	43
11.5 Preparation of study medication.....	43
11.6 Supplies and Accountability.....	44
12. LIST OF REFERENCES.....	45
13. SIGNATURES	49
14. APPENDICES.....	50
APPENDIX 1: Definition of Adverse Drug Events	47
APPENDIX 2: Blood withdrawal and handling of samples	55
APPENDIX 3: Investigator signature page.....	57
APPENDIX 4: Financial statement	58

Abbreviations

AE	Adverse Event
ALAT	Alanine Amino Transferase (also known as SGPT or GPT)
ASAT	Aspartate Amino Transferase (also known as SGOT or GOT)
ATC	Anatomical Therapeutic Chemical (ATC) classification system
AUC	Area Under the Curve
AP	Alkaline Phosphatase
AMG	German Drug Law (Deutsches Arzneimittelgesetz)
BDSG	Bundesdatenschutzgesetz
BfArM	Bundesinstitut für Arzneimittel und Medizinprodukte
CRC	Colorectal Cancer
CRF	Case Report Form
CRP	C-Reactive Protein
CTCAE	Common Toxicity Criteria for Adverse Events
CV	Curriculum Vitae
DBL	Data Base Lock
DFG	Deutsche Forschungsgemeinschaft
DMC	Data Monitoring Committee
DVP	Data Validation Plan
EC	Ethics Committee
ECG	Electrocardiogram
FSI	First Subject In
GCP	Good Clinical Practice
GCP-V	Good Clinical Practice Regulation (GCP-Verordnung)
GGT	Gamma-Glutamyl Transpeptidase
GMP	Good Manufacturing Practice
IB	Investigator's Brochure
iFOBT	Immunochemical Fecal Occult Blood Test
ICH	International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use
IIT	Investigator Initiated Trial
IMP	Investigational Medicinal Product
IMPD	Investigational Medicinal Product Dossier
INN	International Nonproprietary Name
INR	International Normalized Ratio
ISF	Investigator Site File
ISRCTN	International Standard Randomised Controlled Trial Number
ITT	Intention To Treat

KKS	Coordination Center for Clinical Trials (Koordinierungszentrum für Klinische Studien)
LDH	Lactate Dehydrogenase
LKP	Principal/Coordinating Investigator according to AMG (Leiter der Klinischen Prüfung)
LLN	Lower Limit of Normal
LSI	Last Subject In
LSO	Last Subject Out
MCV	Mean Corpuscular Volume
n.a.	Not applicable
PEI	Paul-Ehrlich-Institut
p.o.	per os/ per oral/ orally
ROC	Receiver Operating Characteristics
SAE	Serious Adverse Event
SC	Steering Committee
SDV	Source Data Verification
SGPT	Serum Glutamic-Pyruvate Transaminase (also known as ALAT)
SGOT	Serum Glutamic-Oxaloacetic Transaminase (also known as ASAT)
SmPC	Summary of Product Characteristics
SUSAR	Suspected Unexpected Serious Adverse Reaction
TMF	Trial Master File
ULN	Upper Limit of Normal
WBC	White Blood Cell

1. SUMMARY

Colorectal cancer (CRC) is the third most common cancer and the fourth most common cancer cause of death worldwide, with over 1.2 million new cases and over 600,000 deaths annually. The commonly slow progression from detectable and fully curable precursors (adenomas) to still-curable early-stage CRC to advanced CRC provides much better perspectives for prevention by early detection than for most other cancer types. Thus, the majority of CRCs could be prevented by early detection and removal of adenomas. However, participation of patients in endoscopic screening programs is low, likely due to its invasiveness. Higher participation rates can be achieved with non-invasive stool- or blood-based screening tests. However, such tests suffer from poor sensitivity in detecting precursors for CRC.

In a recent study by our group (Division of Clinical Epidemiology and Aging Research), sensitivity for detecting advanced colorectal neoplasms by immunochemical fecal occult blood test (iFOBT) was 70.8% among users of low-dose acetylsalicylic acid compared with 35.9% among non-users ($p=0.001$), whereas there were only very small differences in specificity. In ROC analysis, the area under the curve (AUC) was much higher for acetylsalicylic acid users than for non-users, with particularly strong differences in men (0.87 versus 0.68, $p=0.003$). This finding suggests that use of acetylsalicylic acid before conduction of iFOBT might be a promising approach to improve non-invasive screening for CRC.

In the randomised, double-blind, placebo-controlled study that we propose, the diagnostic performance (sensitivity, specificity, positive and negative predictive values, likelihood ratios, area under the curve) of two iFOBTs for detecting advanced colorectal neoplasms after a single low dose of acetylsalicylic acid (300 mg) compared to placebo will be evaluated. Acetylsalicylic acid or placebo will be administered at least 5 days before a planned, study-independent colonoscopic screening in 2,400 participants aged 40 to 80 years (i.e. there will be at least 4 full days between drug intake and colonoscopy). Stool samples will be obtained before and on three days after the single dose of acetylsalicylic acid/placebo.

The diagnostic performance of the iFOBTs will be compared to the results of the colonoscopy as a gold standard for the diagnosis of colorectal neoplasms. Additional study questions include gender-specific performance of the tests and gain in diagnostic performance by test application on multiple days. In addition, optional blood samples are taken for future biomarker analyses.

2. KEYWORDS

Colorectal Cancer, Early Detection, Screening, Acetylsalicylic Acid, Sensitivity, Specificity

3. STUDY SYNOPSIS

3.1 Title

Use of Acetylsalicylic Acid (ASA) for Enhanced Early Detection of Colorectal Neoplasms

3.2 Objectives

Main objective

To evaluate diagnostic performance (sensitivity, specificity, positive and negative predictive values, likelihood ratios, area under the curve) of 2 immunochemical Fecal Occult Blood Tests (iFOBTs) for detecting advanced colorectal neoplasms after a single dose of acetylsalicylic acid as compared to placebo.

Secondary objectives

To study gender-specific performance of the 2 iFOBTs and the possible gain in diagnostic performance by stool sampling on multiple days.

To study the safety of single-dose acetylsalicylic acid in the selected population.

To collect blood samples for additional biomarker analyses (optional)

3.3 Intervention

Experimental intervention

Single oral dose of acetylsalicylic acid 300 mg in the form of a tablet without enteric coating.

Control intervention

Single oral dose of a placebo tablet.

Study duration for each participant

5 days (in which day 1 = day on which the study medication is taken). The colonoscopy is not part of the study, but is performed within the framework of the national screening program.

3.4 Endpoints

Primary endpoint

Sensitivity at predefined cutpoints of test positivity of the FOBGold®Tube Screen test to detect advanced neoplasms.

Secondary endpoints

- positive and negative predictive values
- specificity and the area under the curve in ROC analyses
- likelihood ratios
- gender-specific performance of the tests
- gain in diagnostic performance by test application on multiple days
- serious adverse events

3.5 Study type

Multicenter, randomised, double-blind, placebo-controlled study.

3.6 Study phase

This study is an academic, investigator-initiated trial (IIT) using a well known drug. Formally, it could be categorized as a phase III trial, since a potential new indication is analyzed, using a confirmatory approach.

3.7 Study population

Men and women of 40 to 80 years of age with planned screening colonoscopy or planned diagnostic colonoscopy.

3.8 Inclusion and exclusion criteria

Inclusion criteria

- Age 40 to 80 years (both males and females; premenopausal women must have a negative pregnancy test before inclusion into the study, postmenopausal women are women who have not had menstrual bleeding for at least 12 months, or have been surgically sterilized)
- Planned screening or diagnostic colonoscopy
- Able to speak and understand German sufficiently to be able to give written informed consent and comply with the study requirements

Exclusion criteria

- Factors potentially influencing the primary endpoint
 - Diseases/symptoms
 - Chronic inflammatory bowel disease (e.g. Crohn's disease, colitis ulcerosa)
 - Colonoscopy due to positive fecal occult blood test
 - History of Colorectal cancer
 - Angiodysplasia of the colon
 - Anamnestic or observed blood loss per anum
 - Use of any of the following drugs
 - Within 2 weeks before the study
 - Anticoagulants (including, but not limited to heparin, vitamin K antagonists [e.g. phenprocoumon, warfarin], direct thrombin inhibitors [e.g. dabigatran], or factor Xa inhibitors [apixapan, rivaroxaban])
 - Antiplatelet drugs (e.g. clopidogrel, prasugrel, ticlopidin)
 - Within 1 week before the study
 - Acetylsalicylic acid
 - Within 3 days before the study
 - NSAIDs and COX-2 inhibitors
- Factors potentially affecting the safety
 - Any current clinically relevant signs and symptoms, including
 - Signs and symptoms suggesting acute gastrointestinal ulcer
 - Known clinically relevant thrombocytopenia
 - Acute infection

- Volume deficit (exsiccosis)
 - Any currently present allergy with dermal reactions, pruritus, or urticaria
 - Severe or insufficiently controlled asthma
 - Severe kidney or liver diseases (e.g. GFR <30 ml/min, liver cirrhosis)
 - Severe, not sufficiently treated heart failure (as judged by the investigator)
 - Severe, poorly controlled arterial hypertension
 - Any other unclear symptoms needing further investigation in the opinion of the investigator
- Any of the following anamnestic findings
 - History of severe gastrointestinal bleeding
 - Known hemorrhagic diathesis, including, but not limited to, hypoprothrombinaemia, severe thrombocytopenia, hemophilia
 - Asthma, except for patients who have used acetylsalicylic acid in the past without negative effects
 - Hypersensitivity against salicylic acid or other ingredients of the study drugs
 - Previous intolerance to NSAIDs, COX-2 inhibitors, or antirheumatic medication
 - Severe gout (e.g. recurrent attacks)
 - Hereditary oxaluria
 - Known G6PD or glutathione peroxidase deficiency
 - Known epilepsy with generalised seizures
 - Severe cardiac diseases (including, but not limited to, myocardial infarction in the past 6 months)
- Intention to use any of the following drugs during the study
 - Anticoagulants
 - Antiplatelet drugs
 - NSAIDs, COX-2 inhibitors
 - Methotrexate ≥ 15 mg/week
 - Systemic glucocorticoids
 - Selective serotonin reuptake inhibitors (SSRIs)
 - Valproic acid
- Planned surgery/dental treatment during participation in the study
- Other factors
 - Known or suspected relevant alcohol abuse
 - Known or suspected illicit drug abuse
 - Pregnancy
 - Suspected non-compliance with the study procedures
 - Participation in another clinical study

3.9 Statistical analysis

Diagnostic accuracy

Sensitivity, specificity, positive and negative predictive values, likelihood ratios, area under the curve of the two immunochemical FOBTs.

These analyses will be performed for the total study population for whom data on the primary endpoint (sensitivity and specificity for predefined (test-specific) cutpoints of test positivity and the area under the curve in ROC analyses of the FOBGold®Tube Screen Haemoglobin

test on day 3) and where colonoscopy results are available moreover, analyses will be stratified according to gender.

Analyses will be carried out for stool tests taken at baseline before commencing the study medication (if available) and on 3 further days at least 2 days after a single dose of acetylsalicylic acid (or placebo) intake (see study flow chart for further details). Furthermore, analyses will be carried out in which results of tests from stool samples taken on 2 or 3 different days are combined to assess the gain in diagnostic performance by test application on multiple days.

Safety analyses will be done in the total study population (including all randomized patients) using an intention to treat approach.

3.10 Planned sample size

<u>To be randomised:</u>	N= 2,400
<u>To be analyzed (primary endpoint):</u>	N=2,000

3.11 Study duration

Study end is defined as the date when the last patient has completed the study.

Detailed description of trial duration:

Total trial duration:	60 months
Duration of the clinical phase:	48 months
Beginning of the preparation Phase:	Q1 2012/ Jan 2012
FSI (First Subject In):	Q2 2013/ June 2013
LSI (Last Subject In):	Q2 2017/ June 2017
LSO (Last Subject Out):	Q2 2017/ Juni 2017
DBL (Data Base Lock):	Q3 2017/ September 2017
Statistical Analyses Completed:	Q4 2017/ December 2017
Trial Report Completed:	Q1 2018/ March2018

4. BACKGROUND AND RATIONALE

4.1 Background

Colorectal cancer (CRC) is the third most common cancer and the fourth most common cancer cause of death globally, with more than 1.2 million new cases and more than 600,000 deaths annually (1). In Germany, there are over 70,000 new cases and almost 30,000 deaths due to CRC per year (2). The commonly slow progression from detectable and fully curable precursors (adenomas) to still-curable early-stage CRC to advanced CRC provides much better perspectives for prevention by early detection than for most other forms of cancer, and screening for CRC is a highly cost-effective, if not cost-saving, means to reduce the burden of the disease (3).

According to recent evidence from randomized and observational studies, the majority of CRCs (60-90%) could be prevented by endoscopic detection and subsequent removal of adenomas (4, 5). Since October 2002, colonoscopy is offered as a primary screening examination in Germany. However, participation rates in endoscopic screening programs are low, probably due to the invasiveness of the procedure. Higher participation rates can likely be achieved with non-invasive stool- or blood-based screening tests. Randomized studies have demonstrated that CRC incidence and CRC-associated mortality may be reduced by 20-30% with fecal occult blood test (FOBT) screening (6), despite the low sensitivity for detection of colorectal adenomas of conventional (guaiac based) FOBTs. Substantially higher sensitivity for detection of colorectal adenomas can be achieved with newer, immunochemical FOBTs (iFOBTs), but sensitivity for detecting advanced neoplasms still remains below 50% at cut points yielding 90-95% specificity required for population-based screening. Thus, there is an unmet medical need for non-invasive, and, at best, inexpensive and easy-to-implement, screening methods for detection of high-risk colorectal adenomas in addition to CRC.

In a recent prospective screening study conducted among 1979 participants of screening colonoscopy in Germany, sensitivity for detecting advanced colorectal neoplasms by iFOBT was 70.8% among users of low-dose acetylsalicylic acid compared with 35.9% among non-users ($p=0.001$), whereas there were only very small differences in specificity (85.7% for users compared with 89.2 to 91.1% for nonusers) (7). In ROC analysis, the area under the curve (AUC) was much higher among users than among non-users, with differences being particularly pronounced among men (0.87 versus 0.68, $p=0.003$, Figure 1). This AUC for users of low-dose acetylsalicylic acid is the highest AUC of any non-invasive test for detecting advanced colorectal neoplasms reported to date and suggests that use of acetylsalicylic acid prior to conduction of iFOBT might be a particularly promising approach for enhancing non-invasive screening for colorectal cancer. It is hypothesized that the underlying mechanism could be an enhanced tendency of (micro-)bleeding of colorectal neoplasms.

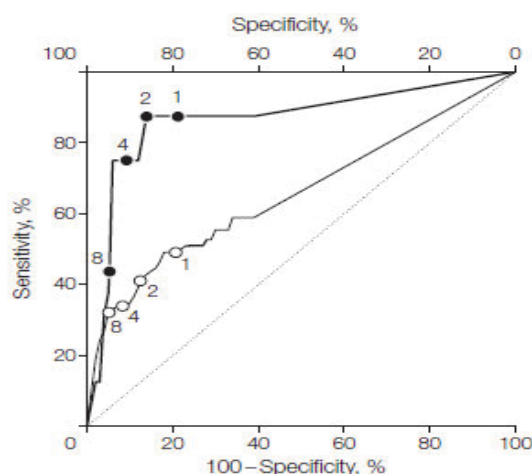


Figure 1: Receiver operating characteristics curve for detecting advanced colorectal neoplasms by quantitative immunochemical fecal occult blood test (RIDASCREEN Hemoglobin) according to use of low-dose acetylsalicylic acid among men (bold line: users, thin line: non-users; circles indicate results for cutpoints 8, 4, 2 and 1 $\mu\text{g per gram of stool}$) From: (7).

4.2 Benefit/ Risk assessment

Acetylsalicylic acid is a very well-known and well-characterized drug that has been on the market for more than 100 years (since 1899). Only one dose is given in this study and this dose (300 mg) is lower than the normal adult dose required for pain relief (500 mg) and is usually classified as “low-dose”. As such it is approved for the secondary prevention of myocardial infarction (where doses from 75 to 325 mg are used). Furthermore, patients with risk factors (e.g. bleeding diathesis) are excluded. Thus, the occurrence of serious adverse reactions is considered very unlikely. More details on the IMP, and possible additional burden for the participants are presented in section 8 (Ethical Considerations).

Previous studies analyzing the association between acetylsalicylic acid use and the performance of FOBTs addressed the potentially hampered specificity due to an increased risk of bleeding from insignificant colonic lesions or from upper gastrointestinal blood loss (8-12). However, these studies mostly used guaiac-based FOBTs, which are more likely to detect upper gastrointestinal blood, as they respond to the pseudoperoxidase activity of the heme moiety of hemoglobin, which remains quite stable throughout the gastrointestinal tract, whereas the iFOBTs respond to globin, which is degraded in the gastrointestinal tract (7-11). A recent observational study on acetylsalicylic acid use and iFOBT performance showed a trend towards an increase in sensitivity (from 51.2% to 66.7%) without loss of specificity in acetylsalicylic acid or NSAID users, but the difference did not reach statistical significance, likely due to a relatively low number of acetylsalicylic acid/NSAID users (10).

Although these results are intriguing, further work is needed for translation of these findings into routine practice of colorectal cancer screening. In particular, low-dose acetylsalicylic acid was used chronically for cardiovascular prevention in most cases and by a relatively small

subgroup of participants (approximately 10%) in our previous observational study. Although knowledge on pharmacokinetics and pharmacodynamics of acetylsalicylic acid suggests similar effects of a single dose of acetylsalicylic acid (as might be considered for colorectal cancer screening) irrespective of cardiovascular risk, this suggestion should be empirically tested in a double-blind placebo-controlled clinical study which we hereby propose. Furthermore, most users of low-dose acetylsalicylic acid in our previous study were men, and data were insufficient to evaluate diagnostic performance among women. Also, our previous study did not allow assessment of potential further improvement by stool testing on multiple days rather than on a single day.

To the best of our knowledge, there has been no previous controlled clinical study investigating the increase in diagnostic accuracy of immunochemical FOBTs with acetylsalicylic acid intake in a large, true target population for colorectal cancer screening, i.e. (largely) asymptomatic women and men at average risk of CRC. If the findings from our recent study are confirmed, this intervention would considerably improve the diagnostic accuracy of immunochemical FOBTs and thus lead to enhanced early detection of colorectal neoplasms. Thus, the results of this study may have a large impact on public health.

5. STUDY DESIGN

The planned study is a randomized, double-blind, placebo-controlled, multicenter study.

Overall, 2,400 eligible participants with an age of 40 to 80 years and with no recent use of acetylsalicylic acid will be recruited into this study when visiting one of the participating gastroenterology private practices or hospitals (outpatient clinics) for an informative appointment, which is routinely scheduled a few weeks before colonoscopy. In addition to participants scheduled for a screening colonoscopy, patients visiting the study centers for a diagnostic colonoscopy but excluding patients with a history of colorectal cancer (see list of in- and exclusion criteria for details) will also be asked to participate. Thus, the colonoscopy procedure is planned and conducted independently from our study.

During this first appointment (day 0), participants will be informed about the study, will receive the printed study information, will have enough time for consideration and will have the opportunity to ask questions. After obtaining written informed consent a detailed medical history and information on the use of medication will be collected and in- and exclusion criteria will be checked. Women who are not postmenopausal have a negative pregnancy test. If in- and exclusion criteria are not met the patient will be considered as a “screening failure” (the reasons will be documented in the patient identification list). If all criteria are met (and the patient consented to the optional blood sampling) four extra tubes of blood (2 EDTA tubes and 2 serum tubes) will be taken for analysis of additional biomarkers (optional). In addition, patients will receive four immunological FOBT-kits for each of two different iFOBTs (FOBGold®Tube Screen, similar test used in our preliminary study (7), as well as the FD Hb/Hp Complex quick test, an internationally well established test (13, 14)(www.frostdiagnostika.de)), so eight kits in total, with instructions on when and how to use them, including a device which is hung in the toilet and aids in easy collection of the stool samples (“Sammelhilfe”). Participants will be randomized to receive either a single dose of 300 mg acetylsalicylic acid or placebo (thereby, a unique participant number is generated and documented). Both participants and their physicians will be blinded with respect to the study medication. Participants will be asked to collect a baseline stool sample on day 0 or on day 1, before taking the study medication. The study medication should be taken on day 1. If the participant cannot collect a stool sample on the first day (e.g. due to severe constipation), this stool sample should be skipped and the study medication should be taken on day 1 without a baseline stool sample. Further stool samples are to be collected from day 3 onwards on 3 different, preferably consecutive days (day 3, 4, and 5). If stool collection on these days is not possible because of constipation or other reasons, stool collection may be postponed to the subsequent day. It is allowed for the participant to give also only two stool samples (before taking study medications and on day 3). At each time point at which stool samples are collected, 1 kit of each test should be used and the actual day documented. The stool sampling devices ensures collection of defined volumes of stool which are given to a buffer that hinders degradation of hemoglobin. All stool samples should be collected at the participants’ homes and prior to initiation of large bowel preparation for colonoscopy and sent by mail to the coordinating center (Heidelberg) using special pre-paid and addressed mailing devices. Participants will receive a form (Participant diary) on which they have to document when they took the study medication, and the date of stool collection. Finally, participants will be asked to fill out a standardized questionnaire addressing potential determinants of risk of colorectal neoplasms and of test performance, including general participant characteristics, co-morbidities, and lifestyle factors. They can either do this during their visit to the study center or at home.

Colonoscopy is planned and conducted independently from our study. However, the investigator will ensure that there are at least 4 acetylsalicylic acid-free days between taking the study medication and the colonoscopy. Although a single dose of acetylsalicylic acid is not considered a contra-indication for elective colonoscopy, this time interval was chosen to minimize any possibility of an increased risk of bleeding at colonoscopy or endoscopic removal of small polyps (see paragraph on safety for further details) (15-17). The colonoscopy is performed in the setting of screening for colorectal cancer or diagnostic workup and as such, it is not part of the study. However, the findings from screening colonoscopy will be collected and used in our analysis.

The maximum time between start of the study and colonoscopy is set at 3 months. At the end of the trial, sensitivity analyses are performed for participants whose colonoscopy is postponed for more than 3 months for any reason. If the colonoscopy does not take place at all, the participant will be considered as a drop-out.

Colonoscopy (and histology) reports will be pseudonymized (i.e. any personal data will be removed and the participant number will be noted) and will be sent from the study center to the coordinating center in Heidelberg following the end of the study for each participant. The relationship between blood samples, stool samples, and colonoscopy results will be established only by the participant number. The information from colonoscopy records will be extracted in a standardized, blinded manner by trained staff.

Blood samples will be picked up by a lab sample transportation service and brought to the coordinating center in Heidelberg. Blood samples will be stored in the coordinating center in Heidelberg for future studies on biomarkers for risk and early detection of advanced adenomas / colorectal cancer including genetic factors. All laboratory analyses will be done in a blinded fashion with respect to both treatment given and clinical/colonoscopy data.

The study is planned and will be conducted according to the principles as stated in the Declaration of Helsinki, applicable local regulations, including German drug law (AMG) and the rules for good clinical practice (ICH-GCP, GCP-V). The study will be initiated only after approval by the Ethics Committee of the Medical Faculty of the University of Heidelberg ("Federführende Ethikkommission"), other responsible ethics committees ("Beteiligte Ethikkommissionen"), and the competent authority (BfArM). Additional study centers will be initiated only after approval by the local ethics committees responsible for the participating study centers. The study will be registered in a clinical trial register before the first patient is included.

An overview of the study flow is shown in Figure 2 and Table 1.

Figure 2 Overview of study flow

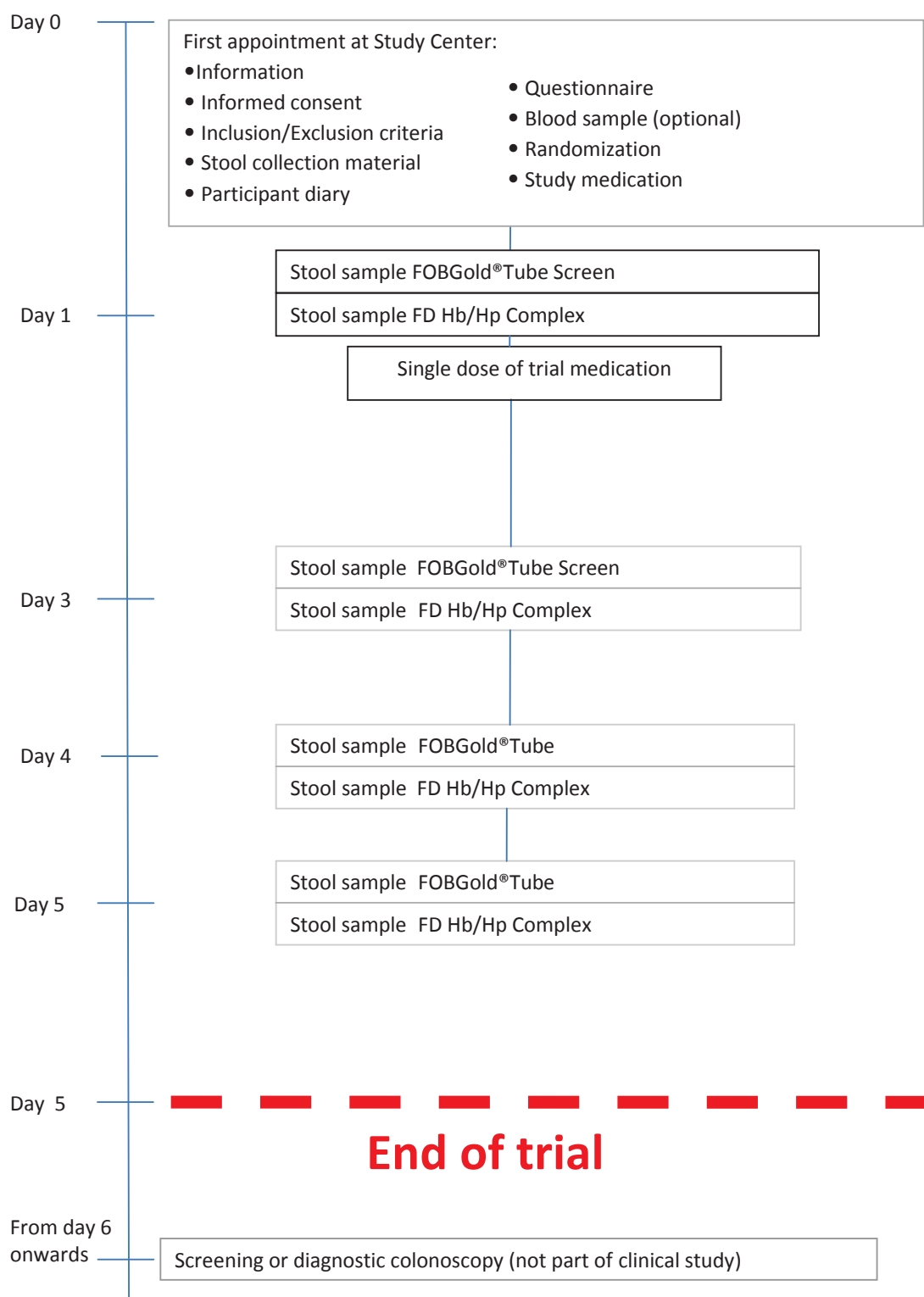


Table 1: Study Schedule

Study specific procedures	Screening	Treatment	Observation				Post-study
			Day 1	Day 3	Day 4	Day 5	
Timeline	0						Colonoscopy from day 6 onwards
Informed consent	X						
In-/ Exclusion criteria	X						
Randomization	X						
Physical examination	X						
Patient parameters ¹	X						
Blood Sampling for Biomarker analysis(optional) ⁵	X						
Questionnaire ²	X						
Handing over stool collection material and subject diary ³	X						
Drug administration (Acetylsalicylic acid or placebo)		X					
Stool Sample FOBGold®Tube Screen ⁴		X	X	X	X		
Stool Sample FD Hb/HpComplex ⁴		X	X	X	X		
Concomitant ther. recording		X	X	X	X		X
(S)AE recording ⁶		X	X	X	X		X

¹ Age, gender, weight, height, and blood pressure

² The questionnaire will include questions on family history of colorectal cancer and lifestyle factors such as smoking and alcohol consumption

³ Eight kits will be handed over

⁴ If stool collection on these days is not possible because of constipation or other reasons, stool collection may be postponed to the subsequent days, it is allowed for the participant to collect also only two stool samples, which should be on days 1 and 3.

⁵ Blood sampling for biomarker analysis: 2 EDTA tubes; 2 serum tubes.

⁶ Interview of subjects before colonoscopy or (if colonoscopy is performed later than 3 weeks after start of study medication) by phone within 3 weeks.

6. DESIGN ASPECTS

6.1 Outcome measures

Primary endpoint

Sensitivity at predefined cutpoints of test positivity of the FOBGold®Tube Screen test to detect advanced neoplasms.

Secondary endpoints

- positive and negative predictive values
- specificity and the area under the curve in ROC analyses
- likelihood ratios
- gender-specific performance of the tests
- gain in diagnostic performance by test application on multiple days
- serious adverse events

6.2 Inclusion / exclusion criteria

Inclusion criteria

- Age 40 to 80 years (both males and females; premenopausal women must have a negative pregnancy test before inclusion into the study, postmenopausal women are women who have not had menstrual bleeding for at least 12 months, or have been surgically sterilized)
- Planned screening or diagnostic colonoscopy
- Able to speak and understand German sufficiently to be able to give written informed consent and comply with the study requirements

Exclusion criteria

- Factors potentially influencing the primary endpoint
 - Diseases/symptoms
 - Chronic inflammatory bowel disease (e.g. Crohn's disease and colitis ulcerosa)
 - Colonoscopy due to positive fecal occult blood test
 - History of Colorectal cancer
 - Angiodysplasia of the colon
 - Anamnestic or observed blood loss per anum
 - Use of any of the following drugs
 - Within 2 weeks before the study
 - Anticoagulants (including, but not limited to heparin, vitamin K antagonists [e.g. phenprocoumon, warfarin], direct thrombin inhibitors [e.g. dabigatran], or factor Xa inhibitors [apixapan, rivaroxaban])
 - Antiplatelet drugs (e.g. clopidogrel, prasugrel, ticlopidin)
 - Within 1 week before the study
 - Acetylsalicylic acid
 - Within 3 days before the study
 - NSAIDs, COX-2 inhibitors

- Factors potentially affecting the safety
 - Any current clinically relevant signs and symptoms, including
 - Signs and symptoms suggesting acute gastrointestinal ulcer
 - Known clinically relevant thrombocytopenia
 - Acute infection
 - Volume deficit (exsiccosis)
 - Any currently present allergy with dermal reactions, pruritus or urticaria.
 - Severe or insufficiently controlled asthma
 - Severe kidney or liver diseases (e.g. GFR <30 ml/min, liver cirrhosis)
 - Severe, not sufficiently treated heart failure (as judged by the investigator)
 - Severe, poorly controlled arterial hypertension
 - Any other unclear symptoms needing further investigation in the opinion of the investigator
 - Any of the following anamnestic findings
 - History of severe gastrointestinal bleeding
 - Known hemorrhagic diathesis, including, but not limited to, hypoprothrombinaemia, severe thrombocytopenia, hemophilia
 - Asthma, except for patients who have used acetylsalicylic acid in the past without negative effects
 - Hypersensitivity against salicylic acid or other ingredients of the study drugs
 - Previous intolerance to NSAIDs, COX-2 inhibitors, or antirheumatic medication
 - Severe gout (e.g. recurrent attacks)
 - Hereditary oxaluria
 - Known G6PD or glutathione peroxidase deficiency
 - Known epilepsy with generalised seizures
 - Severe cardiac diseases (including, but not limited to, myocardial infarction in the past 6 months)
 - Intention to use any of the following drugs during the study
 - Anticoagulants
 - Antiplatelet drugs
 - NSAIDs, COX-2 inhibitors
 - Methotrexate ≥ 15 mg/week
 - Systemic glucocorticoids
 - Selective serotonin reuptake inhibitors (SSRIs)
 - Valproic acid
 - Planned surgery/dental treatment during participation in the study
- Other factors
 - Known or suspected relevant alcohol abuse
 - Known or suspected illicit drug abuse
 - Pregnancy
 - Suspected non-compliance with the study procedures
 - Participation in another clinical study

6.3 Controls / Comparators

Diagnostic performance (sensitivity, specificity, positive and negative predictive values, likelihood ratios, area under the curve) of the two immunochemical FOBTs in participants who received placebo.

6.4 Dose and route of administration

The study medication consists of a single dose of 300 mg acetylsalicylic acid or a placebo to be taken orally.

6.5 Randomization

The randomization list will be prepared after preparation of study drugs. All study drugs will be individually packed and randomly assigned a unique number between 1 and 4,000. A list will be made linking the numbers to the type of study medication given. This list will be kept in a sealed envelope at division of Clinical Epidemiology and Aging Research at the DKFZ. All centres will receive a sufficient number of “blocks”. For each patient to be randomized, the study physician will take the study medication with the next participant number (in ascending order) within the block. This participant number will be documented

- in the CRF
- on the patient identification list
- on stool collection kits
- on blood samples (if applicable)
- on the participant diary
- on the standardized questionnaire
- on the consent form

For each participant number, there will be a sealed envelope with the information on whether it was verum or placebo. One set of these envelopes will be kept at the respective study centers (will be provided together with a “block” of study drugs), one set will be at the KKS and one set will be at the DKFZ. These envelopes are not to be opened at all, except in case of the need for emergency unblinding. In this case, only the envelope corresponding to the study participant for whom unblinding is necessary may be opened. As such, blinding will remain intact for all other participants. Any unblinding must be documented (including the reason for unblinding). During the monitoring visits, the presence and state of the envelopes will be checked. After completion of the study, the sealed envelopes will be returned to the coordinating center, where they will be checked again for completeness and intactness.

If unblinding has taken place for any reason, the study physician will report this to the coordinating center as soon as possible.

6.6 Methods against bias

The study will be randomized to account for potential confounding factors. All laboratory tests as well as extraction of clinical data from colonoscopy and pathology reports will be done in a blinded manner to avoid information bias. Data extraction and data entry (where applicable) will be done by two independent and blinded reviewers. Discrepant coding will be resolved according to standard operating procedures to achieve the maximum accuracy possible.

The study centers as well as all participants will receive detailed instructions to ensure uniform collection and handling of stool and blood samples. Pre-analysis conditions, including data on storage conditions and duration and mode of transportation, will be documented in detail to control for potential variation.

Quality control of data processing will include automated scanning of the questionnaires, optical verification of the scans by trained staff and comprehensive plausibility checks prior to statistical analysis. The staff of Prof. Brenner has extensive expertise with this, based on a large number of ongoing studies in the field conducted by his department.

On-site monitoring will be done by employees of the Coordination Center for Clinical Trials (KKS) Heidelberg and of the Division of Clinical Epidemiology and Aging Research of the German Cancer Research Center.

To ensure participant usage of the study medication, participants will be instructed in detail before start of the study. Given that the medication scheme includes only one single dose of study drug, we do not expect major problems in taking the medication. All participants will receive a participant diary in which they are asked to document if and when they took the study medication. Participants will be asked to bring the empty package of the study medication when coming to the planned colonoscopy. Information on medication will be collected in a drug accountability log.

6.7 Analysis of stool samples

Laboratory analyses of immunochemical FOBGold@Tube Screen test will be done in a blinded manner in the Limbach laboratory, analyses for the FD Hb/HpComplex test will be performed at the laboratory of the Department of Clinical Epidemiology and Aging Research at the German Cancer research Center. Both these laboratory have extensive experience in the analysis of iFOBTs for previous studies of this group. The iFOBTs used in this study are commercially available, validated tests. Standard Operating Procedures will be used for the analyses.

6.8 Analysis and storage of blood samples

Blood samples will be stored at the coordinating center without a time limit for future development and analysis of biomarkers including genetic markers potentially related to the presence of advanced adenomas and/or colorectal carcinoma and for analysis of the determinants of the effects of acetylsalicylic acid. Long-term storage of the blood samples collected in this study will allow evaluating such novel blood markers with minimal delay. As intensive research into new blood tests is ongoing, it is neither useful nor possible to explicitly state which exact markers will be tested in the future. These markers may be based upon the literature, but it is also likely, that markers will be tested that are found in one of the several observational studies into early detection of colorectal cancer that are currently ongoing in the Department of Clinical Epidemiology and Aging Research of the DKFZ. These may include genetic or epigenetic markers. The handling and storage of the blood samples will be conducted in a similar fashion as is currently done in the ongoing observational studies. The blood samples will only be used for research with the specific aims outlined above. Participants will be informed about this and will have the option of participating in the main study without providing a blood sample. In addition, participants will be informed that they can request destroying their blood samples at any time. In that case participants have to contact their study center, which in turn will give the specific participant number to the K357; Aster

EudraCT No.: 2011-005603-32

coordinating center. The coordinating center will ensure that the blood sample will be destroyed. Destruction will be documented.

Participants will not be informed about the results of such analyses, as they have no clinical relevance to them and it is unclear which analyses will be performed at which point in time. Only the staff of the coordinating center will have access to the samples.

The samples will be stored in the coordinating center but may be transported to laboratories of cooperating partners (including international partners) for specific analyses.

6.9 Feasibility of recruitment

To compensate for the time and effort invested and to ensure a high participation rate, the study centers will receive a financial incentive (75€) for each patient recruited. This incentive will be given to the study centers after receipt of the colonoscopy/histology results or after notification by the study center that a participant who had been included and randomized did not undergo colonoscopy within 3 months and the study center has completed the study for this participant according to the study protocol (e.g. documentation of adverse events). In spite of the fact that this financial incentive will be relatively low, we expect that inclusion of 2,400 participants will be feasible, based on our previous experiences in a large ongoing multi-center observational study of early detection of colorectal carcinoma (BliTz Study), in which a smaller financial incentive is given and 4,150 participants from a number of smaller gastroenterology private practices have been included since January 2006. The cooperation structure with the gastroenterology practices that has been built up for this observational study is highly successful, which supports the feasibility of recruiting participants in this manner.

Given the size of the study centers involved in this study and our experience so far, we expect that recruitment of 2,400 participants for this study should be possible within a 48-month period. If recruitment rates are too low, additional study centers may be included.

6.10 Stopping rules

For the individual participant

Participants can discontinue participation in the study at any time without giving a reason. In addition, participants may ask to have their samples destroyed. The investigator will continue to observe patients withdrawn from the study because of intolerable AEs until the findings have been clarified. In all cases where the patient withdraws within 3 weeks, the reason for withdrawal must be recorded in the eCRF (electronic Case Report Form). It is not allowed to destroy data generated in the study. Therefore, in the case of withdrawal of consent the stored data will be checked for further use. Upon request of the participant data not necessary any longer is deleted immediately. Collected personal data will be stored in a pseudonymous manner after completion and publication of the main trial, for at least 10 years, if there are no other regulatory or contractually archiving periods.

The investigator may decide to withdraw a participant, if the participant suffers from a serious adverse event regardless of whether it is drug related or not, if the participant's safety is compromised for any other reason, or if study procedures cannot be completed, if the participant violates the protocol, or any other event occurs that is known to interfere with investigation of the target criteria. This includes the situation in which a participant has taken the study drug too late, so that the time between the day on which the study drug was taken

and the day on which the colonoscopy takes place is less or equal than 4 days. The participant is then withdrawn from the study and the responsible investigator may then decide whether the colonoscopy can take place or not, based on clinical grounds. Alternatively, colonoscopy may be postponed to obtain the required 4-day study-drug free period. In that case, the participant may remain in the study, if there are no other protocol violations.

Participants with a time span between taking the study drug and the colonoscopy of more than 3 months will be documented as protocol violation. At the end of the trial sensitivity analysis are performed..

For the study as a whole

The study will be stopped in case of an inadequate recruitment rate, defined as recruitment of only 25% of participants or less after half of the planned recruiting time and when inclusion of additional study centers appears not promising.

The study will also be terminated in case of an unexpected high rate in serious adverse reactions.

If the study is terminated prematurely, the competent authorities, including the ethics committees and the BfArM, will be informed about this within 15 days.

6.11 Drop-outs

All drop-outs will be recorded by the investigators. Reasonable measures to explore the reasons for dropping out will be employed (e.g. telephone interview) and the reasons documented if the drop-out occurred within 3 weeks. Drop-out individuals cannot be re-included into the study. Their participant numbers will not be re-used.

7. STATISTICAL APPROACH

7.1 Analysis

Standard techniques for the analysis of diagnostic tests will be used (18, 19). Diagnostic accuracy of the two immunochemical FOBTs at the different time points of stool collection will be quantified by estimating sensitivity, specificity, likelihood ratios, positive and negative predictive values (and corresponding 95% confidence intervals) at cut-points yielding levels of specificity typically required for routine testing of average-risk populations (>90%), using the results of colonoscopy as the gold standard. These results will be compared between participants randomized to receive acetylsalicylic acid and those randomized to receive placebo in an intention-to-treat analysis. For the primary endpoint, sensitivity for predefined (test-specific) cutpoints of test positivity of the FOBGold®TubeScreen Haemoglobin test will be compared between users and non-users of acetylsalicylic acid. The stool samples taken on day 3 will be used for these primary analyses.

In addition, the tests will be evaluated over the whole range of cut-points by receiver operating characteristics (ROC) curve analyses and the area under the ROC curve (AUC) will be determined. All analyses will be performed for the total study population, as well as stratified on gender and separately for the different colonoscopy results (advanced adenomas and carcinomas). Analyses will be carried out for stool tests taken at baseline before commencing the study medication and on different days after a single dose of the study drug. Furthermore, analyses will be carried out in which results of tests from stool samples taken on multiple (2 or 3) different days are combined to assess the gain in diagnostic performance by test application on multiple days. Diagnostic performance and its enhancement by acetylsalicylic acid will be evaluated and compared for both of the two iFOBTs employed.

Safety analyses will be performed in the total study population (i.e. all randomized patients, independent from the availability of stool samples or colonoscopy reports).

The primary endpoints will be analyzed with confirmatory aim; the other statistical analyses have an exploratory character. The statistical analysis will be carried out in cooperation with the department of Biostatistics of the DKFZ under guidance and supervision of Prof. Dr. A. Kopp-Schneider, the study statistician.

7.2 Proposed sample size / Power calculations

Diagnostic performance of the test with and without prior use of acetylsalicylic acid will be evaluated using the results of colonoscopy as the gold standard. For the primary endpoint, sensitivity and specificity for predefined (test-specific) cutpoints of test positivity and the area under the curve in ROC analyses of two immunochemical FOBTs taken on day 3 will be compared between users of acetylsalicylic acid and placebo (patients who due to constipation give their first stool sample on day 4 will be excluded from the study). With an expected prevalence of advanced neoplasms of 10%, 100 advanced neoplasms are expected in each of the two groups. Assuming a sensitivity of 36% in the placebo group (as in our preliminary study (7)) and applying 2-sided statistical tests at an alpha level of 0.05 with continuity correction, this study will have a power of 90% to detect an increase in sensitivity by short term use of low-dose acetylsalicylic acid to 60% in the entire study population, and a power of 90% to detect an increase in sensitivity to 70% in gender-specific analyses (sensitivity was 71% in our preliminary study (7)). Furthermore, specificity will be estimated at high levels of

precision in both groups, with an expected confidence interval ranging from 88% to 92% for a specificity of 90%.

8. ETHICAL CONSIDERATIONS/ Risk-Benefit-Evaluation AND SAFETY

8.1 Potential risks of the study medication

Acetylsalicylic acid is a very well-known and well-characterized drug that has been on the market for more than 100 years (since 1899). Only one dose is given in this study and this dose (300 mg) is lower than the normal adult dose required for pain relief (500 mg) and is usually classified as “low-dose”. As such it is approved for the secondary prevention of myocardial infarction (where doses from 75 to 325 mg are used). Furthermore, patients with risk factors (e.g. bleeding diathesis) are excluded. Thus, the occurrence of serious adverse reactions is considered very unlikely.

However, acetylsalicylic acid affects platelet function and as such, it is known to increase the risk of bleeding events, although spontaneous bleeding has been reported to be rare and fatal bleeding events to be exceptional in the setting of primary or secondary prevention of coronary artery disease (20). The absolute risk of upper gastrointestinal events in long-term use of acetylsalicylic acid is estimated to be increased by 0.12% per year of treatment (21). Excess mortality attributable to 1-week use of acetylsalicylic acid is estimated as 185 per 100 million or 0.000185% overall and 0.000417% in people over the age of 60 (22). Thus, although very unlikely, it is theoretically possible that a participant experiences a hemorrhagic event during the study period. Given the short duration of treatment (one dose) and the fact that potential participants with an increased risk of bleeding are excluded, we estimate the risk of such an event to be extremely low.

Candidate-participants in the study will be excluded if they use potentially interacting medication or have illnesses that may be worsened by participation in this study (for details, see exclusion criteria). In the study information documents, participants will be informed about all relevant potential adverse effects.

Before being asked to participate in the study, the participants have all decided to undergo colonoscopy as a screening or diagnostic procedure. The time between drug intake and the planned colonoscopy allows for recovery of haemostatic functions (see below). Thus, the colonoscopy is not considered to be a part of the study.

8.2 Pharmacokinetics and pharmacodynamics of acetylsalicylic acid

Acetylsalicylic acid is eliminated rapidly from the body. The active metabolite, salicylic acid, has a half-life of about 2-3 hours. Thus, one day after administration salicylic acid is eliminated from the body almost completely. However, acetylsalicylic acid-induced platelet dysfunction lasts several days because of the irreversible effects on platelets (i.e. the individual thrombocyte does not recover, but platelet aggregation recovers due to release of newly generated thrombocytes from the bone marrow). As platelet life span is about 10 ± 1 days (23, 24) it is sometimes assumed that the time to normalization of platelet aggregation after administration of acetylsalicylic acid is also 10 days. This assumption appears to be consistent with some laboratory findings where quantified markers (e.g. serum TxB_2 using intensive ex vivo thrombocyte stimulation) normalizes within 2 weeks in most patients (25-28). However, due to experimental conditions, these markers likely reflect the number of functioning platelets but not overall aggregatory function, since complete exchange of thrombocytes is not required to restore normal aggregation. Early on it was recognized that ex

vivo addition of 10-40% acetylsalicylic acid-free platelets apparently normalizes platelet aggregation (29, 30). For example, addition of 40% acetylsalicylic acid-free platelets led to TxB_2 production of 40% (reflecting the number of functioning thrombocytes) but led to aggregation measures of 80% (29). In four human volunteers near-complete functional recovery was earlier (after 5-6.8 days) than individually measured platelet survival time (10.4 days) (25). Based on this data it is estimated that also in vivo 50% functioning thrombocytes will normalize thrombocyte aggregation. Considering the average platelet survival time this will be the case after an average time of 5 days (7 days in individuals with low platelet turn-over). This estimate is in good agreement with a newer and more “physiological” analysis method to quantify aggregation (PFA-100) where aggregation normalized 4-6 days after acetylsalicylic acid administration (31, 32). Furthermore, bleeding time (which is quantified after a defined skin incision) normalized 2-4 days after acetylsalicylic acid administration (31, 33). Finally, colonic mucosal bleeding time was normalized 3 days after acetylsalicylic acid administration (34).

In general, colonoscopy is considered to have a low risk of bleeding complications, with a reported incidence of 0.16% without polypectomy and 0.4-3.4% with polypectomy (35-41). Having taken at least one dose of acetylsalicylic acid within one week before or after polypectomy has been described not to be associated with post-polypectomy bleeding (38, 40), although increased bleeding time of the colonic mucosa shortly after the use of acetylsalicylic acid has been found (34, 42, 43). The current guideline “Endoscopy and antiplatelet agents” of the European Society of Gastrointestinal Endoscopy (44) recommends that for colonoscopy without biopsy sampling, prevalent acetylsalicylic acid use can be continued. Although anticoagulation in general is also a risk factor for post-polypectomy bleeding, the guideline states that acetylsalicylic acid does not increase the risk of post-polypectomy bleeding, irrespective of polyp size. It should be noted, however, that these recommendations were made for patients using acetylsalicylic acid for prevention of cardiovascular events and that the risks of stopping acetylsalicylic acid in these patients were taken into account in making these recommendations. Furthermore, despite the fact that a significant relationship between acetylsalicylic acid use and bleeding risk could not be established so far, the available studies did not have sufficient power to exclude a small increase in bleeding risk. In a recent analysis of 1174 patients with colonoscopic polypectomy the use of ASA or NSAIDs did not increase the risk of postpolypectomy bleeding (45). Similarly, two further studies found no increase in bleeding risk (46, 47).

For other (higher risk) endoscopic procedures the guideline recommends stopping acetylsalicylic acid 5 days before the procedure. To minimize any risk in the present study (where acetylsalicylic acid is given without a clinical indication) a washout phase of at least 4 acetylsalicylic acid-free days before the planned colonoscopy was chosen. If a participant has taken the study drug too late, so that there are less than 4 full days between the day on which the study drug was taken and the day on which the colonoscopy is scheduled, the participant is withdrawn from the study or the colonoscopy postponed. In case of withdrawal, the decision whether the colonoscopy can take place or not then lies with the responsible physician, based on clinical grounds.

8.3 Choice of acetylsalicylic acid dose

Most participants in our previous study (7) used acetylsalicylic acid in a dose of 100 mg per day. This dosing scheme leads to predictable suppression of platelet function after repeated administration. However, the onset of effects is slow and maximal suppression is achieved only after about 5 days (27). Therefore, a loading dose of 500 mg is given in clinical routine when a rapid onset of thrombocyte inhibition is required. Based on published data a dose of 300 mg seems to be similarly effective (48, 49). However, an enteric-coated preparation of the same dose was associated with submaximal inhibition (49) and will therefore not be used in our study.

8.4 Potential benefits of the study medication

The participants in this study have no indication for acetylsalicylic acid and no individual benefit from taking one dose of acetylsalicylic acid. Should this study show that one dose of acetylsalicylic acid strongly increases the sensitivity of the iFOBTs without relevant loss in specificity, these findings may improve diagnostic procedures in future patients, which might affect also study patients if repeated screening is required.

8.5 Safety

Participants will be asked about concurrent use of medication and diseases at inclusion to minimize the risk of adverse reactions. If interacting medications or diseases are present (see exclusion criteria for details), these participants will not be randomized and considered as a “screening failure”. A checklist of in- and exclusion criteria will be part of the electronic Case Report Form (eCRF). Participants will be instructed to contact their general practitioner or the investigator (depending on severity) in the unlikely case of a severe medical problem. The investigator should be informed in any case.

When the participants come for the scheduled study-independent colonoscopy, they will be asked about any potential adverse events or events potentially related to the use of the medication. This interview takes only a few minutes and is performed before colonoscopy (as sedation might be administered during colonoscopy). This information will be documented in a standardized way in the eCRF.

The observation period for adverse events is defined as day 1 to day 5. When colonoscopy is performed later than 3 weeks after taking the study medication patients will be contacted by phone within 3 weeks after taking the study medication in order to evaluate potential adverse events. The investigator follows the outcome of serious adverse events until recovery or stabilization of the participant’s state. Further therapeutic consequences will be documented, for example any medical treatment or drop-out of the study.

The investigator will judge intensity, seriousness and relatedness of an adverse event using standard criteria (see appendix 1 for details). All adverse events (AE) will be documented. All serious adverse events (SAEs) will be reported to the KKS within 24 hours, or on the next working day at the latest, using a standardized SAE-report form. SAE-report forms are checked for inconsistencies, missing data is requested from the investigators, the completed reports are archived and a copy forwarded to the sponsor. A member of the Department of Clinical Pharmacology and Pharmacoepidemiology will act as the medical expert (second assessor) and will check the information on intensity, seriousness, causality and expectedness. All discrepancies with the judgment of the investigator will be discussed and documented. In

case of a suspicion for a SUSAR information on allocation to acetylsalicylic acid or placebo of a specific patient will be provided to the Department of Clinical Pharmacology and Pharmacoepidemiology by the KKS. If placebo has been administered, the medical expert (second assessor) at the Department of Clinical Pharmacology and Pharmacoepidemiology will decide whether the event has to undergo expedited reporting. The information on treatment assignment must not be disclosed to the investigators or other members of the study center.

Suspected unexpected serious adverse reactions (SUSARs) will be reported by the KKS to the responsible ethics committees and the Federal Institute for Drugs and Medical Devices (BfArM) as soon as possible, but in any case within 15 days (7 days in case of a deadly or life-threatening reaction), according to the applicable regulations. The description of the adverse event will be based on CIOMS-I form and as complete as possible. The KKS will inform the responsible investigators of each participating site of all such events. The responsible investigators of the participating sites are responsible for forwarding all such events to all sub-investigators.

Once a year or on request a Development Safety Update Report (DSUR) will be prepared in cooperation between KKS, study coordinator and clinical pharmacology consultant.

If a participant does not undergo the planned colonoscopy for any reason, the participant will be contacted by telephone by the investigator within 3 weeks after taking the study medication to ask for and document the reasons and potential adverse events (if not done before). If the participant agrees, a new colonoscopy appointment may be planned within 3 months after randomization, so that the participant can stay in the trial.

8.6 Emergency unblinding

In case of an emergency situation, in which a treating physician feels that information on whether the participant received acetylsalicylic acid or placebo is needed for adequate medical treatment of the participant, it is possible to unblind the randomization information. To make this possible from the moment of ingestion of the study medication onwards, there will be a sealed envelope for each study participant containing information on the contents of the study drug. One set of these envelopes will be kept at the respective study centers, one set will be at the KKS and one set will be at the DKFZ. If emergency unblinding is necessary, the envelope for that particular study participant may be opened to reveal the information, without unblinding the information for other study participants. The reasons for unblinding will be documented. Whenever possible, the study coordinator should be contacted before the blind is broken.

8.7 Additional burden for the participants

There are no additional visits to the study center for the participants involved in the study. All information, the questionnaire, the material for stool sample collection, and the study drug will be provided during the first visit of the participant. This visit is standard procedure before a screening or diagnostic colonoscopy and not part of the study. For collection of the stool samples, participants will receive a device (one for each day of sample collection), which is hung in the toilet to enable easy collection of stool (Stuhlfänger, Med Auxil). After stool sampling, this device can be flushed in the toilet. Sampling from the stool is done by putting either the supplied small spoon or small stick into the stool and then into a tube, so easy

sampling is ensured. The collected stool samples, the questionnaire and the patient diary can be sent to the coordinating center by normal mail (pre-paid envelopes provided). The (optional) blood samples may be taken at the same time as the routine pre-colonoscopy blood sampling, so there is for most cases no additional puncture involved. Nevertheless, if in any of the participating centers blood withdrawal is not part of the routine pre-colonoscopy examination, patients will be asked if they agree to an additional venous puncture.

Participants will be asked to refrain from using NSAIDs and COX-2 inhibitors during the study. They will be given advice on how to handle pain relief (if necessary) during the study. Paracetamol is the drug of choice if there are no contra-indications. It is assumed that participants who have paracetamol intolerance will be aware of this. Furthermore, patients with severe liver disease or liver cirrhosis are excluded. In addition, patients will be asked not to drink alcohol in excessive amounts during the study.

8.8 Results of the stool tests

As colonoscopy is the gold standard and all participants have already decided to undergo colonoscopy before the study, there is no need to inform participants about the results of their iFOBTs. The colonoscopy is not part of the study and communication of the results of the colonoscopy to the patients is done by the gastroenterologists according to the usual procedure in their practice. However, if a participant partly completes the study, but does not undergo colonoscopy for any reason, the result of the iFOBT may be of importance to him/her. The study center has to inform the coordinating center per fax or email, that the participant did not undergo the scheduled colonoscopy in a period of 3 months after randomization. If the participant agrees (as indicated on the informed consent), a positive stool test result will be sent by the coordinating center in writing to the participant's gastroenterologist if the participant does not undergo colonoscopy in the 3 months following randomization. The gastroenterologist will then be responsible for further steps. If these patients come back after more than 3 months and undergo colonoscopy, they can be reincluded into the trial.

8.9 Results of optional biomarker studies

The blood samples will only be used for research related to colorectal neoplasia. Biomarker analyses may include several markers including genetic or epigenetic markers. Participants will be informed about this and will have the option of participating in the main study without providing a blood sample. In addition, participants will be informed that they can request destroying their blood samples at any time. Participants will not be informed about the results of such analyses, as they have no clinical relevance to them and it is unclear which analyses will be performed at which point in time. Only the staff of the coordinating center will have access to the samples.

In the future, it might be more cost effective to perform whole-exome or even whole-genome analyses instead of analyzing several genetic markers separately. In that case, only markers potentially related to colorectal neoplasia will be extracted from the dataset and analyzed.

9. LEGAL CONSIDERATIONS

9.1 General legal information

The study was planned and will be carried out in accordance with

- a. The legal requirements as laid down in the German “Arzneimittelgesetz (AMG)”, the “Bundesdatenschutzgesetz (BDSG)” and the professional law. The most recent version of the Declaration of Helsinki as adopted by the World Medical Association in October 2008.
- b. The ICH recommendations on Good Clinical Practice (ICH-GCP E 6 R1 2006).
- c. The “Verordnung über die Anwendung der Guten Klinischen Praxis bei der Durchführung von klinischen Prüfungen mit Arzneimitteln zur Anwendung am Menschen des Bundesministeriums für Gesundheit und Soziale Sicherung (GCP-V)” 2004.

9.2 Submission to the Ethics Committees and Competent Authority (BfArM)

The required documents will be submitted to the Ethics Committee of the Medical Faculty of the University of Heidelberg (“federführende Ethikkommission”) and the opinion of the Committee will be obtained in writing. The required documents will also be submitted to the relevant other ethics committees (“beteiligte Ethikkommission”).

The required documents will be submitted to the BfArM and their approval obtained in writing. Participants will not be included until unconditional approval of BfArM and the ethics committees is received in writing.

If the study is terminated as planned, the competent authorities, including the Ethics Committees and the BfArM, will be informed about this within 90 days. In case of unplanned, premature termination, this should be done within 15 days.

Within one year after the end of the study, a clinical study report will be written and sent to the competent authorities, as described in the GCP-V.

9.3 Notification of Authorities

According to § 67 of the German “Arzneimittelgesetz” the local competent authorities will be notified about the study.

9.4 Participant information and informed consent

Before an individual is included in the study, the objectives, risks, and methods of the study will be explained to him or her orally and in writing. The individual gives his/her consent by signing an informed consent form. With their consent, participants agree that their data are given to the competent authorities in pseudonymised form, if necessary. In line with GCP, the participants will receive a copy of the signed and dated informed consent form and the written participant information. If there are any updates to the consent forms or amendments to the written information during the subject’s participation in the study, the subject will be informed about this.

9.5 Responsibilities

Approval of the clinical study protocol by the investigator implies that he agrees to comply with all Good Clinical Practice recommendations as far as relevant for drug studies not performed for drug approval.

9.6 Insurance

According to the legal requirements (§40 AMG) an insurance policy has been issued before the beginning of the study by the ECCLESIA Mildenerger HOSPITAL GmbH.

Policy number: 57 010327 03012.

9.7 Confidentiality

The data obtained in the course of the study will be treated according to the Federal Data Protection Law (Bundesdatenschutzgesetz, BDSG).

During the study, participants will be identified solely by an individual identification code (participant number). Study findings saved on a computer will be stored in accordance with local data protection law and will be handled in the strictest confidence. For protection of these data, organizational procedures are implemented to prevent distribution of data to unauthorized persons. The appropriate regulations of local data will be fulfilled in its entirety.

The participant consents in writing to release the investigator from his/her professional discretion in so far as to allow inspection of original data for monitoring purposes by health authorities and authorized persons. Authorized persons (clinical monitors and inspectors) may inspect the participant-related data collected during the study ensuring the data protection law.

The investigator will maintain a participant identification list (participant numbers with the corresponding participant names) in the study center to enable records to be identified for at least 10 years.

Participants who do not consent to the use of their pseudonymised data will not be included into the study.

In the Informed Consent form, the participants can indicate whether they consent to their gastroenterologist contacting their general practitioner to discuss the results of their stool test if the situation occurs that the participant takes part in the study, but does not undergo colonoscopy and is no longer under the care of the gastroenterologist. If a participant agrees, the gastroenterologist will be released of his obligation of medical confidentiality.

9.8 Archiving

In accordance with the AMG, all documentation containing participant information will be pseudonymized, transferred, and saved for at least 10 years at the coordinating center.

Documentation that cannot be pseudonymized (clinical documentation, patient identification lists, signed consent forms) will be archived in the study centers for at least 10 years.

9.9 Protocol modifications and amendments

If the study protocol has to be changed after approval, a written amendment is required that must be signed by the principal investigator of the study (LKP), the director of the coordinating center, and the clinical pharmacology consultant. Any protocol amendment will only be implemented after approval has been granted by the ethics committees and the competent authorities, where applicable.

Substantial amendments affecting the benefit to risk ratio must be approved by the Ethics Committees and the BfArM. These procedures must be completed before any modifications can come into operation, except when they are necessary to eliminate immediate hazards for the study participants.

9.10 Access to source data, quality control, data handling

For the purpose of ensuring compliance with the Clinical Study Protocol, Good Clinical Practice, and applicable regulatory requirements, the investigator must permit auditing/inspection by applicable regulatory authorities and by representatives of the sponsor (clinical monitors).

The investigator agrees to allow the auditors/inspectors/ monitors to have direct access to the study records for review, being understood that personnel is bound by professional secrecy, and as such will not disclose any personal identity or personal medical information. The investigator will make every effort to help with the performance of the audits, inspections and monitoring, giving access to all necessary facilities, data, and documents.

The participant consents in writing to release the investigator from his/her professional discretion in so far as to allow inspection of original data for monitoring purposes by health authorities and authorized persons. Authorized persons (clinical monitors and inspectors) may inspect the participant-related data collected during the study observing the federal data protection law.

The confidentiality of the data is verified and the protection of the participants should be respected during these inspections.

9.11 Data collection

Electronic CRF (eCRF)

For each participant, an eCRF will be made. Data collected in the eCRF includes:

- Participant number (available once a patient is randomized)
- Participant's medical history and data collected during physical examination
- List of medication taken by the participant within the last 2 weeks before the start of the study.
- Verification of compliance with in- and exclusion criteria
- Dates of participation in the study, including date of informed consent, dispensing of study material, and date of colonoscopy
- Information on whether an optional blood sample was taken
- Laboratory values (thrombocytes) if available from clinical routine
- Adverse events
- Date of drop-out and reasons if drop-out occurred within 3 weeks

The eCRF does not contain the participant's identifying information, but only his/her participant number. The investigator will ensure that the participant's pseudonymity will be maintained. Entries in the eCRF may only be made by the investigator or persons authorized by him. If corrections in the eCRF are necessary, the initial entry will be kept in an electronic log (audit-trail). Potential mistakes found during monitoring visits will be corrected by authorized site staff. Reason for change must be recorded.

A list will be kept of the individuals who are authorized to make data changes.

Informed consent

For each participant in the clinical study written informed consent will be obtained and kept by the investigator in the investigator site file.

Participant diary

All participants will receive a patient diary. In this diary, they shall document if and when they took the study medication, if and when they collected the stool samples. This diary should be sent to the coordinating center in Heidelberg using a prepaid envelop.

List of participants (participants' log, participant identification list)

The responsible investigator of a study center keeps a confidential list with full names, date of birth, and addresses of all participants in the study, giving reference to the participant's records. This list is kept in the investigator site file. For enrolled participants the date of enrolment and the allocation/random number will be recorded. With the help of this list it will be possible to identify the participants and their medical records, e.g. in case of questions arising after completion of the study. The investigator is responsible for ensuring that this list is kept confidential and is stored for 10 years after the end of the study. The identity of the participants will not be revealed to unauthorized persons.

Screening failures will also be documented on this list (without a participation number, which is available only after randomization) together with the reason for failure. Participant log is optional.

9.12 Patient medical records/source data

Patient medical records will be kept for all patients taking part in the study. Participation in the study, administration of study medication, and key data recorded during the study will be documented in the patient medical record as source data unless specified otherwise in this study protocol. Data in the patients' medical records (including original histology reports, colonoscopy reports, etc.) are the 'source data' and entries into the eCRF will be checked against these data (source data verification).

9.13 Drug accountability sheet

The investigator will document on the drug accountability sheet the receipt of study medication, supply of study medication to patients, and return of study medication. All data collected in the study will be reported on paper and kept for ten years.

9.14 Funding

This study is an investigator-initiated trial. For details on the funding of this study, please refer to the financial statement, which is included in appendix 4.

10. QUALITY ASSURANCE AND MONITORING

10.1 Documentation and data handling

All participant data, except data from the participants diary, the questionnaire, colonoscopy and histology reports, will be collected in an electronic Case Report Form (eCRF) by the study physicians and physician assistants on site. All data will be transferred digitally into a database at the DKFZ. Data entries will be checked for inconsistencies. In case of inconsistencies, original reports will be checked for validation.

Participants will be asked to fill out in a participant diary if and when they took the study medication, if and when they collected the stool samples and which other medication (if any) they took during the study. In addition, they are asked to bring the empty package of the study medication or any leftover medication when they come to the study center for the colonoscopy.

The information collected in the diary and from histology and colonoscopy reports will be entered into an additional standardized study database using double entry by two independent and blinded reviewers. Discrepant coding will be resolved according to standard operating procedures to achieve the maximum accuracy possible. Data entries will be checked for inconsistencies through comparison of data sets. In case of differences in data sets, original reports will be checked for validation. Missing data or reports will be requested from the study centers by (e)mail or collected during monitoring visits.

10.2 Monitoring

Monitoring procedures will be adapted to the study-specific risk for the patients, interpretation of ICH-GCP (E6) and GCP-V, and standard operating procedures (SOP) of the KKS to ensure patient safety and the integrity of the clinical data.

Source documents (including pseudonymized colonoscopy reports, histology reports, questionnaires and participant diaries) from the participating study centers will be checked for completeness and plausibility by the monitor at the coordinating center in Heidelberg. The recruitment rate and compliance with eligibility criteria will be supervised and reported back to the study centers, the study coordinator, and the clinical pharmacology consultant.

The participating study centers will be visited regularly, including an initiation visit, a number of interim visits during the recruitment phase and a close-out visit. The number of visits per center (excluding initiation and close-out visits) is estimated at 4 per year, but may vary depending on data quality. Two of these visits will be done by the KKS, the other two will be done by the Division of Clinical Epidemiology and Aging Research of the German Cancer Research Center. During the initiation visit the gastroenterologists and the physician's assistants will be introduced to the study plan, exclusion and inclusion criteria, the design and flow of the study, and evaluation and documentation of adverse events. In addition, they will receive training in how to work with the eCRF. Standardized operating procedures will be presented, explained and discussed in detail. The first monitoring visit by the KKS will be planned relatively early in the study, after inclusion of around 20 patients in a center, to enable early recognition of potential problems or misunderstandings.

Regular on-site monitoring visits, performed by the KKS and the Division of Clinical Epidemiology and Aging Research in an alternating fashion, are planned depending on the K357; Aster

EudraCT No.: 2011-005603-32

recruitment rate and quality of the data. The extent of source data verification and/or the frequency of the monitoring visits will be increased in case of poor quality of the data or regular protocol violations. Frequency of monitoring visits can be reduced in case of high data quality. In addition to the SOPs all procedures will be predefined in a study-specific monitoring manual. Questions about data management (e.g. in case of missing values, implausibility, etc.) will be answered by the investigators in a timely fashion to avoid errors in data capture or entry being carried forward. A monitoring report will be written on each monitoring visit and sent to the sponsor.

10.3 Skills of the Research Physicians

There will be a principal investigator (“Hauptprüfer”) per study center. The principal investigator in each study center will have completed a certified Investigator training, including training in ICH-GCP, GCP-V and AMG, and/or will have a clear track record in conducting clinical drug studies after 2004. Documentation will be collected by the coordinating center and evaluated by the sponsor.

10.4 Steering Committee

There will be a Steering Committee for this study, consisting of the director of the coordinating center, the clinical pharmacology consultant, and the principal investigator of the study (LKP). In addition, the principal investigator from each study center may participate. This committee will meet once before the start of the study and regularly after this, either in person or via a telephone conference, to discuss study progress, monitoring reports, and any study-related problems. The conclusions of these meetings will be documented.

10.5 Publication policy

The results of this study will be written up in one or more scientific paper(s). The aim is to publish the main results in a high-ranking international peer-reviewed medical journal. Writing of all papers will be primarily led by the director of the coordinating center. The study coordinator, the clinical pharmacology consultant, the head of the Department of Clinical Pharmacology and Pharmacoepidemiology, the principal investigator of the study (LKP), and, at least, one investigator of those study centers which contributed substantially to the study will have the possibility to participate. Coauthorship will be based on: 1. Contributions to study concept and design, acquisition of data, analysis and interpretation of the data, 2. Drafting or revising the article for important intellectual content, and 3. Final approval of the version to be published, according to the Uniform Requirements for Manuscripts Submitted to Biomedical Journals (http://www.icmje.org/urm_main.html).

11. STUDY MEDICATION

11.1 General Information about Investigational Medicinal Product

Investigational medicinal product:	Acetylsalicylic acid
International Nonproprietary Name (INN):	Acetylsalicylic acid
ATC code, if officially registered:	N02BA01
Pharmaceutical formulation:	Tablet containing 300 mg
Route of administration:	Oral
Storage conditions:	not above 25°C, protected from humidity
Comparator:	Placebo tablet containing cellulose, lactose-
monohydrate, magnesium stearate	
ATC code, if officially registered:	V03AX
Pharmaceutical formulation:	Tablet
Route of administration:	Oral
Storage conditions:	not above 25°C

Intake of Investigational Medicinal Product

The study drug is administered orally on study day 1 with one glass of water.

11.2 Pharmacological properties

Acetylsalicylic acid is an anti-platelet agent, which inhibits platelet (thrombocyte) aggregation. Its antithrombotic properties are based on the fact that it irreversibly acetylates the enzyme cyclo-oxygenase-1 (COX1) in thrombocytes (platelets), rendering COX1 incapable of converting arachidonic acid to prostaglandin H₂, a precursor of thromboxane A₂. Under normal circumstances, thromboxane A₂ stimulates platelet aggregation and vasoconstriction. Acetylsalicylic acid also acetylates fibrinogen, which plays a role in the stabilization of the thrombus. Low doses of acetylsalicylic acid are used in primary and secondary prevention of vascular complications due to thromboembolic events. As platelets are anuclear, they are incapable of producing new enzymes. Therefore, the inhibition of a platelet lasts for the duration of the life of the platelet and decreases progressively with the generation of new platelets. Per day, 10-15% of platelets are replaced (17, 20, 50, 51). However, aggregatory functions normalize already after approximately 4-6 days, because 50% functioning platelets are sufficient to restore aggregatory functions (as discussed above).

11.3 Pharmacokinetic properties

Acetylsalicylic acid undergoes a first-pass effect in the bowel and the liver by hydrolyzation, leading to the active metabolite salicylic acid. The time to peak concentration in the blood (T_{max}) is 0.5-2 hours. The volume of distribution is 0.16 l/kg. Salicylic acid binds for more than 90% to plasma protein (mostly to albumin). Only the free form has therapeutic potential.

It is further metabolized into inactive metabolites in the liver. Both salicylic acid and its metabolites are excreted in urine, together with a small amount of unmetabolized acetylsalicylic acid (1% of total intake). Elimination half-life is 15-20 minutes for acetylsalicylic acid and 2-3 hours for salicylic acid (52).

11.4 Information about potentially relevant adverse reactions

The risk of adverse reactions due to acetylsalicylic acid is estimated to be low in general and very low in the present study, as patients at higher risk for such complications, such as those with gastrointestinal problems, asthma, known hypersensitivity, and bleeding disorders are excluded (see list of exclusion criteria and discussion above). In addition, the dose given in this study is relatively low and given as a single dose only. Participants who have surgical procedures or dental treatment planned for the study period are excluded. Participants will be informed that if they should need unexpected surgery during the study, there may be an increased risk of bleeding (but that this risk is comparable to that after taking an acetylsalicylic acid containing tablet under normal circumstances).

11.5 Preparation of study medication

The study medication (both acetylsalicylic acid and placebo) will be packaged, labeled and blinded according to applicable GCP-V and GMP rules by the Pharmacy of the University Hospital Heidelberg, holding a manufacturing authorization issued by the Regional Commission of Karlsruhe, Germany on the 31st of January 2012.

The trial medication will be packed by the Pharmacy of the University Hospital Heidelberg according to the randomization list which will also be provided by the Pharmacy (compiled with the randomization software RITA – Randomisation in Treatment Arms, Version 1.24, it.e.c.x).

The tablets will be taken from their original packaging by trained personnel in a clean room class D environment and repackaged and labelled in a blind manner for use in the trial.

Primary packaging: blister with 1 tablet per unit, PVC/PVDC blisterfoil for adequate vapour barrier, white-opaque, sealed to aluminium blisterfoil, blind labelling, no secondary packaging

The trial medication will be labelled according to § 5 of GCP-V. The label will contain the following details:

- For use in clinical trial only
- Name, address of the trial sponsor [German Cancer Research Center, Im Neuenheimer Feld 672, 69120 Heidelberg] or his/ her representative (e.g. CRO),
- Phone number of the sponsor [in case of Investigator Initiated Trial (IIT) phone number of the coordinating/ principal investigator]¹
- Name and potency of investigational medicinal product
- Batch number
- Dosing instructions or a reference to a separate document/ instructions of investigator

- Administration form
- Content specified by weight, volume or quantity
- Application form²
- Expiry date using note “usable until”
- Code of Trial Protocol¹
- EudraCT-Number ⁴
- Subject’s Identification Code
- Instructions on required storage conditions
- Keep inaccessible for children³

[notice: ¹ not applicable, if included in an accompanying document to be handed out to the trial subject, ² not applicable for firm connected immediate and outer packaging and for oral formulations, ³ only applicable, if the trial medication is to be handed out to the trial subjects, ⁴ not applicable, if included in an accompanying document]

After completion of the study or after the period required by the applicable regulatory authorities, whichever represents the longer retention period, any leftover medication will be destroyed at the study centers. Date, type and amount of medication and way of destruction will be documented in the drug accountability log.

11.6 Supplies and Accountability

The investigator will keep an account of the trial medication and acknowledge the receipt of all shipments of the trial medication. All trial medication must be kept in a locked area with access restricted to designated trial staff. The trial medication must be stored in accordance with manufacturer’s instructions at below 25°C and protected from humidity. The investigator will also keep accurate records of the quantities of trial medication dispensed, used, and returned by each subject. The documentation has to include date of dispensary, subject identification, batch/ serial numbers or other identification of trial medication. The site monitor will periodically check the supplies of trial medication held by the investigator to ensure the correct accountability of all trial medication used. At the end of the trial, all unused trial medication and all medication containers will be completely returned to the Pharmacy of the University Hospital Heidelberg where it will be destroyed, documentation of drug destruction will be provided. It will be assured that a final report of the drug accountability is prepared and maintained by the investigator.

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13. SIGNATURES

The present trial protocol was subject to critical review and has been approved in the present version by the persons undersigned. The information contained is consistent with:

- the current risk-benefit assessment of the investigational medicinal product
- the moral, ethical, and scientific principles governing clinical research as set out in the applicable version of Declaration of Helsinki and the principles of GCP.

The investigator will be supplied with details of any significant or new finding including AEs relating to treatment with the investigational medicinal product.

Date: 11.8.15

Signature:

Name (block letters): Prof. Dr. med. Thomas Seufferlein

Function: Principal Investigator of the Study (LKP)

Date: 24.7.2015

Signature:

Name (block letters): Prof. Dr. med. Brenner

Function: Director of the Coordinating Center

Date: 27.7.2015

Signature:

Name (block letters): PD Dr. med. Czock

Function: Clinical Pharmacology Consultant

Date: 24.7.2015

Signature:

Name (block letters): Prof. Dr. Kornel Schneider

Function: Biometrician

13. SIGNATURES

The present trial protocol was subject to critical review and has been approved in the present version by the persons undersigned. The information contained is consistent with:

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Date: _____ Signature: _____
Name (block letters): Prof. Dr. med. Thomas Seufferlein
Function: Principal Investigator of the Study (LKP)

Date: _____ Signature: _____
Name (block letters): Prof. Dr. med. Brenner
Function: Director of the Coordinating Center

Date: _____ Signature: _____
Name (block letters): PD Dr. med. Czock
Function: Clinical Pharmacology Consultant

Date: _____ Signature: _____
Name (block letters): Prof. Dr. Kopp-Schneider
Function: Biometrician

14. APPENDICES

APPENDIX 1: Definition of Adverse Drug Events

Definition of an Adverse Event (AE)

According to GCP, an adverse event (AE) is defined as follows: Any untoward medical occurrence in a subject administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

An AE may be:

- New symptoms/ medical conditions
- New diagnosis
- Changes of laboratory parameters
- Intercurrent diseases and accidents
- Worsening of medical conditions/ diseases existing before clinical trial start
- Recurrence of disease
- Increase of frequency or intensity of episodic diseases.

A pre-existing disease or symptom will not be considered an adverse event unless there will be an untoward change in its intensity, frequency or quality. This change will be documented by an investigator.

Surgical procedures themselves are not AEs; they are therapeutic measures for conditions that require surgery. The condition for which the surgery is required may be an AE. Planned surgical measures permitted by the clinical trial protocol and the condition(s) leading to these measures are not AEs, if the condition leading to the measure was present prior to inclusion into the trial.

AEs are classified as "non-serious" or "serious".

Definition of a Serious Adverse Event (SAE)

A serious adverse event is an adverse event that at any time during the period of observation:

- results in death
- is life-threatening
- requires or prolongs hospitalization
- results in persistent or significant disability or incapacity
- involves congenital anomaly

Definition of Unexpected Adverse Event

An unexpected adverse event is an adverse event of which the nature and severity is not consistent with the applicable product information.

Furthermore, reports which add significant information on specificity, severity or frequency of a known adverse reaction constitute 'unexpected' events.

Definition of Suspected Unexpected Serious Adverse Reaction (SUSAR)

A SUSAR is a suspected unexpected serious adverse reaction. All adverse events that are suspected to be related to an investigational medicinal product and that are both unexpected and serious are considered to be SUSARs.

Intensity

The assessment of intensity will be based on the investigator's clinical judgment using the following definitions:

- Mild: an event that is easily tolerated by the patient, causing minimal discomfort and not interfering with everyday activities.
- Moderate: an event that is sufficiently discomforting to interfere with normal everyday activities.
- Severe: An event that prevents normal everyday activities.

Causality

The causal relationship of an adverse event to treatment (treatment indicates all investigational medicinal products including placebo) has to be assessed in accordance with WHO criteria in a modified version: Edwards IR, Biriell C. Harmonisation in Pharmacovigilance. Drug Safety 2004;10:93-102.

The investigator will use clinical judgment to determine the relationship. Alternative causes, such as natural history of the underlying diseases, concomitant therapy, other risk factors etc. will be considered. The investigator will also consult the Investigator Brochure or other product information. The relationship between the drug and the occurrence of each adverse event will be evaluated and classified as:

Certain: A clinical event, including laboratory test abnormality, occurs in a plausible time relationship to drug administration, and which cannot be explained by concurrent disease or other drugs or chemicals. The response to withdrawal of the drug (dechallenge) should be clinically plausible. The event must be definitive pharmacologically or phenomenologically, using a satisfactory rechallenge procedure if necessary.

Probable: A clinical event, including laboratory test abnormality, with a reasonable time sequence to administration of the drug, unlikely to be attributed to concurrent disease or other drugs or chemicals, and which follows a clinically reasonable response on withdrawal (dechallenge). Rechallenge information is not required to fulfill this definition.

Possible: A clinical event, including laboratory test abnormality, with a reasonable time sequence to administration of the drug, but which could also be explained by concurrent disease or other drugs or chemicals. Information on drug withdrawal may be lacking or unclear.

Unlikely: A clinical event, including laboratory test abnormality, with a temporal relationship to drug administration which makes a causal relationship improbable, and in which other drugs, chemicals or underlying disease provide plausible explanations.

Not assessable: Reports which for one or another reason are not yet assessable, e.g. because of outstanding information (can only be a temporary assessment).

Not related: A clinical event, including laboratory test abnormality, with no relationship to drug administration, and in which other drugs, chemicals or circumstances provide plausible explanations.

Reporting of Serious Adverse Events by Investigator

All SAEs must be reported by the investigator to the responsible Safety Officer (KKS) within 24 hours after becoming aware of the SAE, or at the latest on the next working day, using the "Serious Adverse Event" form.

The initial report must be as complete as possible including details of the current illness and (serious) adverse event and an assessment of the causal relationship between the event and the study medication. (The investigator must also inform the site monitor in all cases).

Expedited Reporting

SUSARs are to be reported to the ethics committee(s), competent higher federal authority (BfArM) and to all participating principal investigators within regulative defined timelines, i.e. they are subject to an expedited reporting. All principal investigators of each participating site are obliged to inform their sub-investigators about SUSARs.

Investigators participating in this trial will report all SAEs to a responsible Safety Officer at the KKS Heidelberg as soon as possible but not later than 24 hours, or at the latest on the next working day, after their notification. The reporting will be performed by faxing of a completed 'SAE Form' to the KKS Heidelberg, fax-number: 06221-56-33725.

All SAE will be subject to a second assessment by a designated person, who will be independent from the reporting investigator, the trial sponsor and the KKS Heidelberg. The designated person for the present trial, referred to as the second assessor is the clinical pharmacology consultant or another qualified person from the department of clinical pharmacology and pharmacoepidemiology.

The second assessor will fill out a 'Second Assessment Form' for each SAE and send it back per fax to the responsible person at the KKS Heidelberg within the next 3 working days (72 hours), fax-number:

06221-56-33725

The 'Second Assessment Form' will contain the following information:

- I) assessment of relationship between SAE and IMP
- II) assessment of expectedness of SAE (derived from IB or SmPC)
- III) statement if the benefit/ risk assessment for the trial did change as a result of SAE.

The expedited reporting will be carried out by a responsible Safety Officer at KKS Heidelberg. If applicable, the Safety Officer will break the blind. Only SUSARs occurring after administration of IMP will undergo expedited reporting.

In the case of SUSARs that resulted in death or were life-threatening the competent authority, the Ethics Committee, and all principal investigators must be notified as soon as possible but no later than 7 calendar days after the safety officer has first knowledge of the minimum criteria for expedited reporting. In each case relevant follow-up information should be sought and a report completed as soon as possible. It should be communicated to the competent authority and the ethics committees within an additional eight calendar days. All other SUSARs and safety issues must be reported to the competent authority and the ethics committees as soon as possible but no later than 15 calendar days after safety officer has first knowledge of the minimum criteria for expedited reporting. Further relevant follow-up information should be given as soon as possible. Person-related data have to be pseudonymised by using a participant code number.

Each SUSAR has to be reported as a single case to the competent authority and the ethics committee. Expedited reporting is not usually required for reactions which are serious but expected. In case of a SUSAR after administration of placebo a decision regarding the need of expedited reporting will be taken by the medical expert (second assessor).

For regulatory purposes, initial expedited reports should be submitted within the time limits as soon as the minimum following criteria are met:

- A suspected investigational medicinal product
- An identifiable participant (pseudonymised)
- An adverse event assessed as serious and unexpected, and for which there is a reasonable suspected causal relationship
- An identifiable reporting source

Outcome of an adverse event

The outcome of an AE at the time of the last observation will be classified as:

‘Recovered/resolved’: all signs and symptoms of an AE disappeared without any sequels at the time of the last interrogation.

‘Recovering/resolving’: the intensity of signs and symptoms has been diminishing and/or their clinical pattern has been changing up to the time of the last interrogation in a way typical for its resolution.

‘Not recovered/not resolved’: signs and symptoms of an AE are mostly unchanged at the time of the last interrogation.

‘Recovered/resolved with sequel’: actual signs and symptoms of an AE disappeared but there are sequels related to AE.

‘Fatal’: resulting in death. If there are more than one adverse event only the adverse event leading to death (possibly related) will be characterized as “fatal”.

‘Unknown’: the outcome is unknown or implausible and the information cannot be supplemented or verified.

Action taken with IMP

This term refers to the specific actions taken to treat or alleviate adverse events or to avoid their sequels. Following categories will be used to define the countermeasures to adverse events.

The action taken with IMP will be assigned to one of the following categories:

‘Unknown’: the information is unknown or implausible and it cannot be supplemented or verified.

‘Not applicable’: the question is implausible (e.g. the participant is dead.)

None: no action taken.

Drug treatment: newly prescribed medication or change in dose of a medication

Others: other countermeasures, e.g. a surgical procedure

References

1. Detailed guidance on the European database of Suspected Unexpected Serious Adverse Reactions Eudravigilance – Clinical Trial Module, April 2004
2. Verordnung über die Anwendung der Guten Klinischen Praxis bei der Durchführung von klinischen Prüfungen mit Arzneimitteln zur Anwendung am Menschen (GCP Verordnung-GCP-V), Bundesgesetzblatt Jahrgang 2004 Teil I Nr. 42. 9. August 2004

3. Clinical Safety Data Management: Data elements for transmission of individual case safety reports, CPMP/ICH/287/95 The European Agency for the Evaluation of Medicinal products, London, 16. November 2000

4. Note for Guidance on Definitions and Standards for expedited reporting (CPMP/ICH/3945/03) The European Agency for the Evaluation of Medicinal products, London, 20. November 2003

APPENDIX 2: Blood withdrawal and handling of samples

Blood withdrawal

During the first appointment (day 0), if all inclusion and exclusion criteria are met (and the patient consented to the optional blood sampling) four extra tubes of blood 9-10ml each (2 EDTA tubes and 2 serum tubes) will be taken for analysis of additional biomarkers (optional). Nevertheless, if in any of the participating centers blood withdrawal is not part of the routine pre-colonoscopy examination, patients will be asked if they agree to an additional venous puncture.

Tubes used for blood collection:

8.5 to 9ml serum will be collected using either of the following tubes depending on the center:

- Sarstedt S-Monovette 9 ml (Type.-Nr. 02.1388)
- Vacutainer SST II Serum tubes, 8.5 ml (Type.-Nr. 367953)

9 to 10ml EDTA tubes for blood collection using either of the following tubes depending on the center:

- Sarstedt S-Monovette, 9 ml, (Type.-Nr. 06.1664.001), Kalium-EDTA (K3)
- Vacutainer K2E, 10 ml, (Type.-Nr. 367525), Kalium-EDTA (K2)

Handling and storage of blood samples

After blood withdrawal, serum samples will be kept at room temperature for 30 minutes and for a maximum of 60 minutes, to allow complete coagulation of the serum sample. Followed by 10 minutes centrifugation at 2500-3500 RPM. Blood samples (EDTA tubes) will be gently shaken (turned upside down) 8 to 10 times to prevent coagulation. Both serum and plasma samples will be stored at the centers at 4°C to 8°C until being picked up by the Limbach Laboratory within 2 to 3 days of blood withdrawal. Blood samples will be transported (on ice). At the laboratory of the Division of Clinical Epidemiology and Aging Research (C070) blood samples will be centrifuged (2500-3500 RPM) and aliquoted on the same day. After centrifugation the buffy coat (white blood cells) will be collected from the plasma samples. Aliquots from both serum and plasma will be stored at -20°C.

Blood samples may also be transported to laboratories of cooperating partners (including international partners) for future studies on biomarkers for risk and early detection of advanced adenomas / colorectal cancer including genetic factors. All laboratory analyses will be done in a blinded fashion with respect to both treatment given and clinical/colonoscopy data.

The handling and storage of the blood samples will be conducted in a similar fashion as is currently done in the ongoing observational studies. The blood samples will only be used for research with the specific aims outlined above. Participants will be informed about this and will have the option of participating in the main study without providing a blood sample. In addition, participants will be informed that they can request destroying their blood samples at any time. In that case participants have to contact their study center, which in turn will give the specific participant number to the coordinating center. Participants will not be informed about the results of such analyses, as they have no clinical relevance to them and it is unclear

which analyses will be performed at which point in time. Only the staff of the coordinating center will have access to the samples.

Involved laboratories

Ulrike Schlesselmann

German Cancer Research Center (Storage and analysis of blood and stool samples)

Division of Clinical Epidemiology and Aging Research (C070)

Im Neuenheimer Feld 581

D-69120 Heidelberg

Phone: +496221421310

u.schlesselmann@dkfz-heidelberg.de

Herr. Heinz Jürgen Roth

Labor Limbach (Transportation and aliquotting of blood samples)

Im Breitspiel 15

D-69126 Heidelberg

Phone: +49 3432-511

roth@labor-limbach.de

APPENDIX 3: Investigator signature page

Protocol Number: K357

Protocol Title: Use of acetylsalicylic acid for enhanced early detection of colorectal neoplasms

I agree:

- to assume responsibility for the proper conduct of the study at this site.
- to conduct the study in compliance with this protocol, any future amendments, and with any other study conduct procedures provided by the sponsor.
- not to implement any deviations from or changes to the protocol without agreement from the sponsor and prior review and written approval from the ethics committees, except where necessary to eliminate an immediate hazard to the subjects.
- that I am thoroughly familiar with the appropriate use of the investigational drug, as described in this protocol, and any other information provided by the sponsor including, but not limited to, the approved product label.
- that I am aware of and will comply with current good clinical practices guidelines (GCP-V, ICH-GCP) and all regulatory requirements.
- to ensure that all persons assisting me with the study are adequately informed about the investigational drug and their study-related duties and function as described in the protocol.
- to comply with procedures for data recording/reporting
- to give direct access to all study related sites, source data/documents and reports for the purpose of monitoring, auditing, and regulatory inspections
- to retain the study related essential documents until the sponsor informs me that these documents are no longer needed

Date, place :

Printed name

Signature

.....

.....

Use of acetylsalicylic acid (ASA) for enhanced early detection of colorectal neoplasms (CONFIDENTIAL)
Study protocol.

Version 1.5; 24-July-2015

APPENDIX 4: Financial statement

Studientitel: Mit ASS Darmtumoren früher erkennen?
Kurztitel: K357

Ersteller:
Datum:

Kostenpunkt	Kosten	Finanziert durch	Aufgabenwahrnehmung über*	
	[Euro]		Studieninitiiierenden Prüfer bzw. zugehöriges Zentrum	delegiert an
1- Projektmanagement (KKS)**	13,383	Abteilung Klinische Epidemiologie und Altersforschung (C070)		KKS
Studienprotokoll	4,424			
Patienten Information	1,580			
Versicherung	632			
Antrag auf zustimmende Bewertung bei der EK ¹	3,476			
Antrag auf Genehmigung bei der BOB	2,528			
Landesbehörde	395			
Studienabschluss	348			
2- Projektmanagement (intern)**	153,000	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
3- Projektmanagement (KlinPharm)**	30,950	Abteilung Klinische Pharmakologie und Pharmakoepidemiologie		KlinPharm
Erstellung Studienprotokoll	8,000			
Erstellung Patienten Information	2,000			
Antrag auf zustimmende Bewertung bei EK und BOB	400			
Prüfung Qualifikation Studienzentren und Prüfärzte	650			
Beurteilung von SAEs und SUSARs	10,000			
Prüfung von Amendments	800			
Prüfung von Monitoring Reports und DSURs	3,200			
Steering Committee Treffen	300			
Mitverfassen des Abschlussberichts	5,600			
4- Qualitätskontrolle: Monitoring^{3**}	24,790			KKS
Prüfartzordner	1,776			
Monitoring Manual	1,184			
Einarbeitung und Training	1,184			
Initiierungs-Visit	2,131			
Regelmäßiges Monitoring	14,208		Abteilung C070	KKS
Close out visits	1,421			

K357; Aster

EudraCT No.: 2011-005603-32

Use of acetylsalicylic acid (ASA) for enhanced early detection of colorectal neoplasms (CONFIDENTIAL)
Study protocol.

Version 1.5; 24-July-2015

Zentrenbetreuung	2,886			
5- Reisekosten für Monitoring⁴	3,000		Abteilung C070	KKS
6- Pharmakovigilanz**	28,922	Abteilung Klinische Epidemiologie und Altersforschung (C070)	KlinPharm	KKS
Erstellung des Safety-Teils des Protokolls	884			
Safety manual	2,272			
Erfassung und Meldung von SAE und SUSARS ⁵	15,600			
Safety reports	8,616			
Erstellung von SAE-Listings in der klinischen Phase	1.704			
Pflege der Pharmakovigilanz Strukturen	1,548			
7- Datenmanagement (inkl.eCRF)**	40,642	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	KKS
Datenmanagementplan	504			
Erstellen der Datenbank	7,288			
Kodierung von unerwünschten Ereignissen	nz			
Statusberichte	189			
Queryprozess	15,309			
Datenbankschluß	1,008			
Datentransfer	1,008			
MACRO-Lizenz	10,800			
MACRO-Support	4,536			
8- Allgemeine Projektkosten (KKS)**	12,731	Abteilung Klinische Epidemiologie und Altersforschung (C070)		KKS
Studientreffen	2,592			
Zentraler Studienordner	5,573			
Koordination/Kommunikation	4,566			
Qualitätssicherung: Audits	nz			
9- Aufwandsentschädigung für Patienten Praxen⁶	60,000	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
10- Studienmedikation/Verblistern	3,400	Apotheke Universitätsklinikum Heidelberg / Im Neuenheimer Feld 670 69120		
11- Prüfarzt training/GCP⁷	2,430			
12- Labor**	24,881	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	Labor Limbach
Kurierkosten ⁸	1,440			
Laborbedarf ⁹	2,388			

Use of acetylsalicylic acid (ASA) for enhanced early detection of colorectal neoplasms (CONFIDENTIAL)
Study protocol. Version 1.5; 24-July-2015

Materialkosten ¹⁰	6,107			
Versandkosten ¹¹	546			
Messinstrumente/ -methoden ¹²	14,400			
13- Sonstige Treffen¹³	500	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
14- Gebühren (EK, BfArM)¹⁴	6,800	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
15- Versicherung¹⁵	67,116	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
16- Allgemeine Kosten **	35,060	Abteilung Klinische Epidemiologie und Altersforschung (C070)	Abteilung C070	
Druckaufträge ¹⁶	3,000			
Versandkosten ¹⁷	31,080			
Tagung und Hotel ¹⁸	980			
Studienbudget	505,570			
Overhead (OH)				
Mehrwertsteuer (MWST)				
Summe aus Studienbudget, MWST, OH				
vertraglich eingeworbene Mittel				
Mittel aus eigenem Budget				
Summe zur Verfügung stehende Mittel				

**Gesamtsumme der aus den folgenden Punkten entstehenden Kosten

¹Einschließlich der Registrierung bei der federführenden EK (2528€) und bei weiteren beteiligten EK (948€)

²Gehalt TV-L E13 (50%) für 3 Jahre (30.000€*3) und Gehalt TV-L E9 (50%) für 3 Jahre (21.000€*3). Für Projektkoordination, Data Management, Monitoring, Statistische Auswertungen

³Das Monitoring wird nach ICH E6 (GCP), entsprechend den SOPs des KKS Heidelberg und dem studien-spezifischen Monitoring Manual durchgeführt

⁴KKS: 2 Besuche pro Jahr über 3 Jahre, Zugticket 80€+ 53€ pro Stunde + möglichen Übernachtungskosten (80€)=1596€.
Für Benseid and Zaineddin: 2 Besuche pro Jahr über 3 Jahre, 80€ Zugticket pro Person=480€. Gesamtkosten (minimum) 2076€

⁵Angenommen wird: 1 von 20 Teilnehmern hat eine SAE (100*150€) und 1 von 1000 hat eine SUSAR (2*300€)

⁶30€ für jeden abgeschlossenen Fall (30*2000€)

⁷Prüfarzttraining für 3 Ärzte 650€*3=1950€, plus Übernachtung 80€*3=240€, plus bahn 80€*3=240€= 2430€

⁸Kurierkosten (z.B. für Laborproben, Medikation) Blutproben abholen: 0,3€/Probe*2 Proben*2400=1440€

⁹2 Plasmaröhrchen und 2 Serumröhrchen pro Person. 100 Plasmaröhrchen kosten 27,7€, 100 Serumröhrchen kosten 26,6€

¹⁰Materialkosten (z.B. Schutzhöhrchen 8 Schutzhöhrchen pro Person, für 2400 Teilnehmer, 500 Schutzhöhrchen kosten 133€, plus 1000€ Stuhlauffanghilfe

¹¹3,5€ pro Lieferung an das DKFZ jeden Montag/52Wochen*3 Jahre=546€

¹²1,5€ pro FOBGold@TubeScreen Messung* 4 Proben je Teilnehmer*2400=14.400€

Use of acetylsalicylic acid (ASA) for enhanced early detection of colorectal neoplasms (CONFIDENTIAL)
Study protocol. Version 1.5; 24-July-2015

¹³Reisekosten und Verpflegung

¹⁴EK (Ethikkommission), BfArM (Bundesinstitut für Arzneimittel und Medizinprodukte). BfArM 1700€ für die Anmeldung der Studie sowie 1000€ für jeden Jahresbericht, 700€ pro Amendment (Angenommen 3 Amendments)

¹⁵23.50€/Person*2400

¹⁶Druckkosten für 5000 Flyer liegen bei 350-500€, Druckkosten für 3000 Kopien des Fragebogens betragen durchschnittlich 1045€ (ohne MwSt.) . Die Druckkosten für alle Studiendokumente liegen bei 3500€. Dazu können weitere Kosten für Farbdruke kommen.

¹⁷Briefporto 1.45€*2400= 3480€, Versandkosten für Stuhlproben 2.25€*4*2400= 21.600€, plus Versand von Dokumenten von den Praxen 2400*2.50€ =6000€, gesamt 31080€

¹⁸ 1 mal pro Jahr Studienmeeting in Heidelberg (insgesamt 4 Meetings), Hotelkosten (80€), Reisekosten von Ulm/HD (80€) für 3 Personen= 480€. Plus weiterer mögliche Kosten von 500€.